Note for Integrating Factors:

We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example, $e^{\int \frac{1}{x} dx} = e^{\ln|x|} = |x|$

On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1) a)

Inflow is constant

$\therefore \frac{dV}{dt} = k$

1) b)

Leak rate is proportional to the volume of water present

Outflow $= kV$

$\therefore \frac{dV}{dt} = -kV$

1) c)

Inflow $= 6$

Leak rate is proportional to the volume of water present

Outflow $= kx$

$\therefore \frac{dx}{dt} = 6 - kx$

1) d)

Rate of change is proportional to $(200 - V)$

$\therefore \frac{dx}{dt} = k(200 - V)$

1) e)

Cooling at a rate proportional to $(T - 18)$

$\therefore \frac{dT}{dt} = -k(T - 18)$

1) f)

Inflow $= 0.5t$

Leak rate is proportional to the amount of pollutant present

$\therefore \frac{\mathrm{d}P}{\mathrm{d}t} = 0.5t - kP$

2)

Driving Force: $(18 - 3t)$ N

Resistive Force: $4v$ N

Mass $= 6$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Using $F = ma$:

$(18 - 3t) - 4v = 6\frac{\mathrm{d}v}{\mathrm{d}t}$

Or $\frac{\mathrm{d}v}{\mathrm{d}t} = 3 - \frac{1}{2}t - \frac{2}{3}v$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{3}v = 3 - \frac{1}{2}t$

3)

Pulling Force: 25 N

Resistive Force: $(5 + 0.2t)v$ N

Mass $= 10$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Using $F = ma$:

$25 - (5 + 0.2t)v = 10\frac{\mathrm{d}v}{\mathrm{d}t}$

Or $\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{5}{2} - \frac{5+0.2t}{10}v$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{25+t}{50}v = \frac{5}{2}$

4)

Taking downward as positive.

Force acting downwards is just the weight: 80g N

Force acting upwards is air resistance: $6v$ N

Mass $= 80$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Using $F = ma$:

$80\text{g} - 6v = 80\frac{\mathrm{d}v}{\mathrm{d}t}$

Or $\frac{\mathrm{d}v}{\mathrm{d}t} = \text{g} - \frac{3}{40}v$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{3}{40}v = \text{g}$

5) a)

Driving Force: 6e^{-t} N

Resistive Force: v^2 N

Mass $= 2$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Using $F = ma$:

$6\text{e}^{-t} - v^2 = 2\frac{\mathrm{d}v}{\mathrm{d}t}$

Or $\frac{\mathrm{d}v}{\mathrm{d}t} = 3\text{e}^{-t} - \frac{1}{2}v^2$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{2}v^2 = 3\text{e}^{-t}$

5) b)

The differential equation is not linear because it contains v^2, so the integrating factor method cannot be applied.

6) a)

Forward Force: 8 N

Resistive Force: $2v\frac{\mathrm{d}v}{\mathrm{d}t}$ N

Mass $= 5$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Using $F = ma$:

$8 - 2v\frac{\mathrm{d}v}{\mathrm{d}t} = 5\frac{\mathrm{d}v}{\mathrm{d}t}$

Or $(5 + 2v)\frac{\mathrm{d}v}{\mathrm{d}t} = 8$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{8}{5+2v}$

6) b)

The differential equation is not linear because it contains $\frac{8}{5+2v}$ and cannot be rearranged into the form $\frac{\mathrm{d}v}{\mathrm{d}t} + \mathrm{P}(t)v = \mathrm{Q}(t)$, so the integrating factor method cannot be applied.

6) c)

Use Separation of Variables

$\int (5 + 2v)\,\mathrm{d}v = \int 8\,\mathrm{d}t$

$\Rightarrow 5v + v^2 = 8t + c$

Substitute $v = 0.5$ and $t = 0$:

$5 \times 0.5 + 0.5^2 = 8 \times 0 + c \Rightarrow c = 2.75$

$\therefore 5v + v^2 = 8t + 2.75$

$\Rightarrow (v + 2.5)^2 - 6.25 = 8t + 2.75$

$\Rightarrow (v + 2.5)^2 = 8t + 9$

$\Rightarrow v + 2.5 = \pm\sqrt{8t + 9}$

$\Rightarrow v = -2.5 \pm \sqrt{8t + 9}$

As the initial velocity is 0.5 ms^{-1}, we must have $v = -2.5 + \sqrt{8t + 9}$

7) a)

Concentration $= \frac{12}{80} = 0.15$ kg / l

7) b)

Concentration $= \frac{S}{V}$ kg / l

7) c)

Concentration $= \frac{5}{200} = 0.025$ kg / l

Salt leaves at $3 \times 0.025 = 0.075$ kg / min

8)

Concentration $= \frac{S}{100}$ kg / l

Salt leaves at $4 \times \frac{S}{100} = \frac{S}{25}$ kg / min

9) a)

Salt enters the tank at $5 \times 0.3 = 1.5$ kg / min

9) b)

Concentration $= \frac{S}{100}$ kg / l

Salt leaves the tank at $5 \times \frac{S}{100} = \frac{S}{20}$ kg / min

9) c)

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 1.5 - \frac{S}{20}$

10)

Salt enters the tank at $2 \times 0.4 = 0.8$ kg / min

Salt leaves the tank at $2 \times \frac{S}{50} = \frac{S}{25}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 0.8 - \frac{S}{25}$

Where $S(0) = 2$

11)

Net change in volume $= 3 - 5 = -2$

$\therefore V = 100 - 2t$

12)

Net change in volume $= 7 - 2 = 5$

$\therefore V = 60 + 5t$

Where $t \geq 0$

13)

Net change in volume $= p - q$

$\therefore V = V_0 + (p - q)t$

14)

No salt enters the tank (pure water) $= 0$ kg / min

Salt leaves the tank at $4 \times \frac{S}{100} = \frac{S}{25}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = -\frac{S}{25}$

15) a)

Net change in volume $= 4 - 6 = -2$

$\therefore V = 100 - 2t$

15) b)

Salt leaves the tank at $6 \times \frac{S}{100-2t} = \frac{6S}{100-2t}$ kg / min

15) c)

No salt enters the tank (pure water) $= 0$ kg / min

Salt leaves the tank at $\frac{6S}{100-2t}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = -\frac{6S}{100-2t}$

16)

Net change in volume $= 4 - 6 = -2$

$\therefore V = 100 - 2t$

Valid for $0 \leq t \leq 50$

Salt enters the tank at $4 \times 0.2 = 0.8$ kg / min

Salt leaves the tank at $6 \times \frac{S}{100-2t} = \frac{6S}{100-2t}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 0.8 - \frac{6S}{100-2t}$

17) a)

Net change in volume $= 3 - 2 = 1$

$\therefore V = 40 + t$

17) b)

Salt enters the tank at $3 \times 0.5 = 1.5$ kg / min

Salt leaves the tank at $2 \times \frac{S}{40+t} = \frac{2S}{40+t}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 1.5 - \frac{2S}{40+t}$

17) c)

The incoming concentration of salt in the water is 0.5 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so the concentration tends towards 0.5 kg per litre

18) a)

The number 18 represents room temperature, so $18℃$.

18) b)

$(T - 18)$ is the difference between the tea's temperature and $18℃$, so it represents how far the tea is above room temperature.

18) c)

If $T > 18$, then $T - 18 > 0$, so $\frac{\mathrm{d}T}{\mathrm{d}t} < 0$, meaning the tea is cooling down. The negative sign ensures the temperature decreases when the tea is hotter than the room.

18) d)

0.12 controls how quickly cooling happens. A larger value would mean faster cooling. Its units are per minute.

19) a)

$0.4P$ represents population growth proportional to the current population. In this context, as $0.4 > 0$, it would represent births.

19) b)

-2 represents a constant decrease in the population, such as a fixed number of fish being removed each year (e.g. fishing or harvesting)

19) c)

-2 does not depend on P as the model assumes the same number of fish are removed each year regardless of how many fish are in the lake. i.e. harvesting is assumed to be constant (not proportional)

20) a)

The term 6 is a constant positive value, so it represents a constant charge in coulombs per second.

20) b)

The term $\frac{Q}{4}$ increases as Q increases, so it represents an effect that becomes stronger the more charge is already stored. Since it is subtracted, it represents a factor that reduces the charging rate. So $\frac{Q}{4}$ represents a slowing effect that increases as the capacitor fills up.

20) c)

As Q increases, $\frac{Q}{4}$ increases, so the value of $6 - \frac{Q}{4}$ decreases.

So the charging rate decreases as the capacitor stores more charge, which matches the realistic idea that it becomes harder to add more charge once it is already highly charged. So the negative term ensures the charging slows down as Q increases.

20) d)

As Q becomes large, $\frac{Q}{4}$ becomes large, so $6 - \frac{Q}{4}$ decreases.

Eventually, when $6 - \frac{Q}{4} = 0 \Rightarrow Q = 24$

This means the capacitor approaches a maximum charge of 24 C and the charging rate $\frac{\mathrm{d}Q}{\mathrm{d}t}$ tends to zero.

21) a)

The equation is of the form $\frac{\mathrm{d}S}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

So 0.8 is the rate at which salt enters the tank

i.e. 0.8 kg per minute

21) b)

$\frac{5S}{100}$ must be the rate at which salt leaves the tank

21) c)

In the $\frac{5S}{100}$ term, we can infer that because there is a constant denominator, the volume of liquid in the tank is remaining constant (at 100 litres).

It is structurally of the form $5 \times \frac{S}{100}$, so liquid is exiting and entering the tank at 5 litres per minute.

21) d)

The inflow concentration is $\frac{0.8 \text{ kg per min}}{5 \text{ litres per min}} = 0.16$ kg per litre

22) a)

The equation is of the form $\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

So 3.5 is the rate at which oxygen enters the tank

i.e. 3.5 cm^3 per hour

22) b)

The equation is of the form $\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

So since the term $\frac{2x}{50+4t}$ is being subtracted, it represents the rate at which oxygen is leaving the tank.

22) c)

$-0.5 = 3.5 - 2 \times \frac{150}{50+4t}$

$\Rightarrow \frac{150}{50+4t} = 2$

$\Rightarrow \frac{75}{25+2t} = 2$

$\Rightarrow 75 = 50 + 4t$

$\Rightarrow t = 6.25$ hours

$\Rightarrow t = 375$ minutes

23) a)

$\frac{\mathrm{d}S}{\mathrm{d}t}$ means 'rate of change of S with respect to time'

Since S is measured in kg and t is measured in minutes, the units are kg per minute

23) b)

Each term in the equation must have the same units as $\frac{\mathrm{d}S}{\mathrm{d}t}$, so 0.6 must also be measured in kg per minute, and represents the rate of salt entering the tank.

23) c)

Rewrite $\frac{3S}{40}$ as $3 \times \frac{S}{40}$

S is measured in kg

In a mixing problem, the 3 represents the outflow with units litres per minute.

The 40 represents a constant volume, so 40 has units of litres.

So $3 \times \frac{S}{40}$ has units $(\text{litres per minute}) \times \frac{\text{kg}}{\text{litres}} = \text{kg per minute}$ which matches $\frac{\mathrm{d}S}{\mathrm{d}t}$

24) a)

$\frac{\mathrm{d}x}{\mathrm{d}t}$ is the rate of change of mass with respect to time, so $\frac{\mathrm{d}x}{\mathrm{d}t}$ has units grams per minute. So the constant 1.5 must also have the units grams per minute.

24) b)

$\frac{4x}{80+2t}$ is the outflow term.

x is measured in grams.

$80 + 2t$ is measured in litres (representing the volume at time t)

So $\frac{x}{80+2t}$ has units grams per litre

$\frac{4x}{80+2t}$ must have units grams per minute (to be consistent with $\frac{\mathrm{d}x}{\mathrm{d}t}$)

So 4 must have units litres per minute (representing the outflow rate)

24) c)

If x were measured in kg, then $\frac{\mathrm{d}x}{\mathrm{d}t}$ would have units kg per minute. But the constant term 1.5 was interpreted as 1.5 grams per minute. So the left hand side would be kg per minute and the right hand side would be in grams per minute. This would make the equation dimensionally inconsistent.

The constant term could be adjusted:

1.5 grams per minute $= 0.0015$ kg per minute

25) a)

$\frac{\mathrm{d}P}{\mathrm{d}t}$ has units thousands of bacteria per hour

25) b)

Each term in the differential equation must match the units of $\frac{\mathrm{d}P}{\mathrm{d}t}$ so k must also have units thousands of bacteria per hour

25) c)

t is measured in hours, so $12 + t$ must be measured in hours, therefore the number 12 must represent 12 hours.

25) d)

$t = \frac{T}{60} \Rightarrow \frac{\mathrm{d}t}{\mathrm{d}T} = \frac{1}{60}$

$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\mathrm{d}P}{\mathrm{d}t} \times \frac{\mathrm{d}t}{\mathrm{d}T}$

$\Rightarrow \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{1}{60}\frac{\mathrm{d}P}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{1}{60}\left(k - \frac{P}{12+t}\right)$

$\Rightarrow \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{1}{60}\left(k - \frac{P}{12+\frac{T}{60}}\right)$

$\Rightarrow \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{1}{60}\left(k - \frac{60P}{720+T}\right)$

$\Rightarrow \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{k}{60} - \frac{P}{720+T}$

26) a)

The volume decreases by 3 litres per minute, so:

$V = 90 - 3t$

26) b)

$90 - 3t \geq 0 \Rightarrow t \leq 30$

So $0 \leq t \leq 30$ is the domain for which the model is physically valid.

26) c)

For $t > 30$, the formula gives $V < 0$, which is impossible as volume cannot be negative. The model is not valid for $t < 0$ as the situation is defined from $t = 0$.

27) a)

There are 6 litres flowing into the tank per minute, and 2 litres flowing out of the tank per minute, so the net inflow rate is 4 litres per minute:

$V(t) = 40 + 6t - 2t$

$\Rightarrow V(t) = 40 + 4t$

27) b)

$120 = 40 + 4t$

$\Rightarrow t = 20$

So the tank starts to overflow after 20 minutes.

27) c)

The model assumes the volume continues increasing, but this stops being true when the tank overflows. So the valid domain is:

$0 \leq t \leq 20$

27) d)

The model assumes flow rates in and out of the tank are constant.

28) a)

Use Separation of Variables

$\frac{\mathrm{d}P}{\mathrm{d}t} = 0.12P$

$\Rightarrow \int \frac{1}{P}\,\mathrm{d}P = \int 0.12\,\mathrm{d}t$

$\Rightarrow \ln|P| = 0.12t + c$

$\Rightarrow P = \pm \mathrm{e}^{0.12t+c}$

$\Rightarrow P = A\mathrm{e}^{0.12t}$

When $t = 0$, $P = 500$:

$500 = A\mathrm{e}^{0} \Rightarrow A = 500$

$\therefore P = 500\mathrm{e}^{0.12t}$

28) b)

$P|_{t=20} = 500\mathrm{e}^{0.12\times 20}$

$= 5511.58819\ldots$

$= 5510$ (3 SF)

28) c)

This is exponential growth, which assumes unlimited food, space and no disease. In reality, as the population grows, resources become limited, so the growth rate decreases.

28) d)

A better model would reduce the growth rate as the population becomes large, to reflect limited resources. One example would be $\frac{\mathrm{d}P}{\mathrm{d}t} = 0.12P - kP^2$

29) a)

If $T > 20$ then $T - 20 > 0$ so $\frac{\mathrm{d}T}{\mathrm{d}t} < 0$, which implies the temperature decreases.

If $T < 20$ then $T - 20 < 0$ so $\frac{\mathrm{d}T}{\mathrm{d}t} > 0$, which implies the temperature increases.

So T approaches 20℃ but does not go below it.

29) b)

Use Separation of Variables

$\frac{\mathrm{d}T}{\mathrm{d}t} = -0.18(T - 20)$

$\Rightarrow \int \frac{1}{T-20}\,\mathrm{d}T = \int -0.18\,\mathrm{d}t$

$\Rightarrow \ln|T - 20| = -0.18t + c$

$\Rightarrow T - 20 = \pm \mathrm{e}^{-0.18t+c}$

$\Rightarrow M = 20 \pm \mathrm{e}^{-0.18t+c}$

$\Rightarrow M = 20 + A\mathrm{e}^{-0.18t}$

When $t = 0$, $T = 70$:

$70 = 20 + A\mathrm{e}^{-0.18\times 0} \Rightarrow A = 50$

$\therefore M = 20 + 50\mathrm{e}^{-0.18t}$

29) c)

Assumptions include:

- The room temperature stays constant at 20℃
- The coffee is well-mixed so the temperature is uniform throughout the liquid.
- Heat loss is proportional to the difference in the temperature.

30) a)

The equation contains $100 - t$ in the denominator, so we must have that $t \neq 100$. So as t is time, we must have $t \geq 0$, and hence:

$0 \leq t < 100$

30) b)

The model is undefined at $t = 100$, since we have $100 - t$ in the denominator. Also, for $t > 100$, the denominator is negative, so $\frac{\mathrm{d}M}{\mathrm{d}t} > 0$, meaning the mass would increase rather than decay. Hence the model cannot be valid for all $t \geq 0$

30) c)

Use Separation of Variables

$\frac{\mathrm{d}M}{\mathrm{d}t} = -\frac{5M}{100-t}$

$\Rightarrow \int \frac{1}{M}\mathrm{d}M = \int -\frac{5}{100-t}\mathrm{d}t$

$\Rightarrow \ln|M| = 5\ln|100 - t| + c$

$\Rightarrow \ln|M| = \ln|(100 - t)^5| + c$

$\Rightarrow M = \pm \mathrm{e}^{\ln|(100-t)^5|+c}$

$\Rightarrow M = A(100 - t)^5$

When $t = 0$, $M = 80$:

$80 = A(100 - 0)^5 \Rightarrow A = \frac{80}{100^5}$

$\therefore M = \frac{80}{100^5} \times (100 - t)^5$

$\Rightarrow M = \frac{80(100-t)^5}{100^5}$

30) d)

The model is valid for $0 \leq t < 100$, so we consider t getting close to 100 but still less than 100. As $t \to 100$, $100 - t \to 0$

So $M = A(100 - t)^5 \to 0$

Therefore the mass tends to zero.

30) e)

The model suggests that the chemical always reaches exactly zero mass at $t = 100$, which is unlikely in reality. Also the decay rate becomes infinite as $t \to 100$, which is physically impossible.

31) a)

Net change in volume $= 4 - 4 = 0$

$\therefore V = 120$

Salt enters the tank at $4 \times 0.25 = 1$ kg / min

Salt leaves the tank at $4 \times \frac{S}{120} = \frac{S}{30}$ kg / min

$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 1 - \frac{S}{30}$ as required

31) b)

$\frac{\mathrm{d}S}{\mathrm{d}t} = 1 - \frac{S}{30}$

First, rearrange into the form $\frac{\mathrm{d}S}{\mathrm{d}t} + P(t)S = Q(t)$:

$\frac{\mathrm{d}S}{\mathrm{d}t} + \frac{S}{30} = 1$

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{30}\mathrm{d}t}$

$= \mathrm{e}^{\frac{1}{30}t}$

$\frac{\mathrm{d}S}{\mathrm{d}t} + \frac{S}{30} = 1$

$\Rightarrow \mathrm{e}^{\frac{1}{30}t}\frac{\mathrm{d}S}{\mathrm{d}t} + \mathrm{e}^{\frac{1}{30}t}\frac{S}{30} = \mathrm{e}^{\frac{1}{30}t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{e}^{\frac{1}{30}t}S\right) = \mathrm{e}^{\frac{1}{30}t}$

$\Rightarrow \mathrm{e}^{\frac{1}{30}t}S = 30\mathrm{e}^{\frac{1}{30}t} + c$

$\Rightarrow S = 30 + c\mathrm{e}^{-\frac{1}{30}t}$

Substitute $S = 10$ and $t = 0$:

$10 = 30 + c\mathrm{e}^{-\frac{1}{30}\times 0}$

$\Rightarrow 10 = 30 + c$

$\Rightarrow c = -20$

$\therefore S = 30 - 20\mathrm{e}^{-\frac{1}{30}t}$

31) c)

$S|_{t=20} = 30 - 20\mathrm{e}^{-\frac{1}{30}\times 20}$
$= 19.7$ kg (3 SF)

32) a)

Net change in volume $= 8 - 6 = 2$
$\therefore V = 50 + 2t$
Salt enters the tank at $8 \times 0.25 = 2$ kg / min
Salt leaves the tank at $6 \times \frac{S}{50+2t} = \frac{6S}{50+2t} = \frac{3S}{25+t}$ kg / min
$\frac{\mathrm{d}S}{\mathrm{d}t} = \text{in} - \text{out}$
$\therefore \frac{\mathrm{d}S}{\mathrm{d}t} = 2 - \frac{3S}{25+t}$ as required

32) b)

$\frac{\mathrm{d}S}{\mathrm{d}t} = 2 - \frac{3S}{25+t}$

First, rearrange into the form $\frac{\mathrm{d}S}{\mathrm{d}t} + \mathrm{P}(t)S = \mathrm{Q}(t)$:

$\frac{\mathrm{d}S}{\mathrm{d}t} + \frac{3S}{25+t} = 2$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{25+t}\mathrm{d}t}$

$= \mathrm{e}^{3\ln|25+t|}$

$= \mathrm{e}^{\ln|(25+t)^3|}$

$= |(25+t)^3|$

Let Integrating Factor $= (25+t)^3$

$\frac{\mathrm{d}S}{\mathrm{d}t} + \frac{3S}{25+t} = 2$

$\Rightarrow (25+t)^3 \frac{\mathrm{d}S}{\mathrm{d}t} + \frac{3S}{25+t} \times (25+t)^3 = 2(25+t)^3$

$\Rightarrow (25+t)^3 \frac{\mathrm{d}S}{\mathrm{d}t} + 3S(25+t)^2 = 2(25+t)^3$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(S(25+t)^3) = 2(25+t)^3$

$\Rightarrow S(25+t)^3 = \frac{1}{2}(25+t)^4 + c$

$\Rightarrow S = \frac{1}{2}(25+t) + c(25+t)^{-3}$

Substitute $S = 5.5$ and $t = 0$:

$5.5 = \frac{1}{2}(25+0) + c(25+0)^{-3}$

$\Rightarrow 5.5 = \frac{25}{2} + \frac{c}{15625}$

$\Rightarrow c = -109375$

$\therefore S = \frac{1}{2}(25+t) - 109375(25+t)^{-3}$

$\Rightarrow S = \frac{1}{2}(25+t) - \frac{109375}{(25+t)^3}$

32) c)

The incoming concentration of salt in the water is 0.25 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so the concentration tends towards 0.25 kg per litre

Alternatively:

The concentration is $\frac{S}{V} = \frac{S}{50+2t} = \frac{S}{2(25+t)}$

$= \frac{\frac{1}{2}(25+t) - \frac{109375}{(25+t)^3}}{2(25+t)}$

$= \frac{1}{4} - \frac{109375}{2(25+t)^4}$

As $t \to \infty$, $\frac{109375}{2(25+t)^4} \to 0$ so the concentration tends towards $\frac{1}{4} = 0.25$ kg per litre

32) d)

The model assumes that the salt is perfectly mixed with the water, so the concentration is uniform throughout the tank at all times.

33) a)

Net change in volume $= 70 - 30 = 40$

$\therefore V = 600 + 40t$

Carbon dioxide enters the vessel at $70 \times 0.08 = 5.6$ cm^3 / hour

Carbon dioxide leaves the vessel at $30 \times \frac{x}{600+40t} = \frac{30x}{600+40t} = \frac{3x}{60+4t}$ cm^3 / hour

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}x}{\mathrm{d}t} = 5.6 - \frac{3x}{60+4t}$ as required

33) b)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 5.6 - \frac{3x}{60+4t}$

First, rearrange into the form $\frac{\mathrm{d}x}{\mathrm{d}t} + \mathrm{P}(t)x = \mathrm{Q}(t)$:

$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60+4t} = 5.6$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{60+4t}\mathrm{d}t}$

$= \mathrm{e}^{\frac{3}{4}\ln|60+4t|}$

$= \mathrm{e}^{\ln\left|(60+4t)^{\frac{3}{4}}\right|}$

$= \left|(60+4t)^{\frac{3}{4}}\right|$

Let Integrating Factor $= (60+4t)^{\frac{3}{4}}$

$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60+4t} = 5.6$

$\Rightarrow (60+4t)^{\frac{3}{4}}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60+4t} \times (60+4t)^{\frac{3}{4}} = 5.6(60+4t)^{\frac{3}{4}}$

$\Rightarrow (60+4t)^{\frac{3}{4}}\frac{\mathrm{d}x}{\mathrm{d}t} + 3x(60+4t)^{-\frac{1}{4}} = 5.6(60+4t)^{\frac{3}{4}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(x(60+4t)^{\frac{3}{4}}\right) = 5.6(60+4t)^{\frac{3}{4}}$

$\Rightarrow x(60+4t)^{\frac{3}{4}} = 0.8(60+4t)^{\frac{7}{4}} + c$

$\Rightarrow x = 0.8(60+4t) + c(60+4t)^{-\frac{3}{4}}$

Initially the vessel only contains nitrogen, so no carbon dioxide

Substitute $x = 0$ and $t = 0$:

$0 = 0.8(60 + 4 \times 0) + c(60 + 4 \times 0)^{-\frac{3}{4}}$

$\Rightarrow 0 = 48 + 60^{-\frac{3}{4}}c$

$\Rightarrow c = -48 \times 60^{\frac{3}{4}}$

$\therefore x = 0.8(60+4t) - 48 \times 60^{\frac{3}{4}} \times (60+4t)^{-\frac{3}{4}}$

$\Rightarrow x = 0.8(60+4t) - 48\left(\frac{60}{60+4t}\right)^{\frac{3}{4}}$

$x|_{t=5} = 0.8(60 + 4 \times 5) - 48\left(\frac{60}{60+4\times5}\right)^{\frac{3}{4}}$

$= 64 - 38.684\ldots$

$= 25.3\ \text{cm}^3$ (3 SF)

33) c)

The incoming concentration of salt in the water is 0.25 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so the concentration tends towards 0.25 kg per litre

Alternatively:

The concentration is $\frac{x}{V} = \frac{x}{600+40t} = \frac{x}{10(60+4t)}$

$= \frac{0.8(60+4t) - 48\left(\frac{60}{60+4t}\right)^{\frac{3}{4}}}{10(60+4t)}$

$= 0.08 - \frac{48}{10} \times \frac{60^{\frac{3}{4}}}{(60+4t)^{\frac{7}{4}}}$

As $t \to \infty$, $\frac{48}{10} \times \frac{60^{\frac{3}{4}}}{(60+4t)^{\frac{7}{4}}} \to 0$ so the concentration tends towards $0.08 = 8\%$

33) d)

Possible answers include:

- The gases may not mix instantly and uniformly throughout the vessel.
- The inflow and leak rates may not remain constant as the pressure changes.
- The vessel may not be able to keep increasing in total volume indefinitely.
- The percentage of carbon dioxide in the incoming mixture may not remain exactly constant.

34) a)

Net change in volume $= 3 + 1 - 4 = 0$

Volume of ethanol and water in the vessel is constant

$\therefore V = 15 + 20 = 40$ litres

The fraction of ethanol in the vessel is $\frac{x}{40}$

Ethanol leaves the vessel at $4 \times \frac{x}{40} = \frac{x}{10}$ litres / second

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{in} - \text{out}$

$\therefore \frac{\mathrm{d}x}{\mathrm{d}t} = 3 - \frac{x}{10}$

So $k = 10$

34) b)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3 - \frac{x}{10}$

First, rearrange into the form $\frac{\mathrm{d}x}{\mathrm{d}t} + \mathrm{P}(t)x = \mathrm{Q}(t)$:

$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{x}{10} = 3$

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{10}\mathrm{d}t}$

$= \mathrm{e}^{\frac{1}{10}t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{x}{10} = 3$

$\Rightarrow \mathrm{e}^{\frac{1}{10}t}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{x}{10}\mathrm{e}^{\frac{1}{10}t} = 3\mathrm{e}^{\frac{1}{10}t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(x\mathrm{e}^{\frac{1}{10}t}\right) = 3\mathrm{e}^{\frac{1}{10}t}$

$\Rightarrow x\mathrm{e}^{\frac{1}{10}t} = 30\mathrm{e}^{\frac{1}{10}t} + c$

$\Rightarrow x = 30 + c\mathrm{e}^{-\frac{1}{10}t}$

Initially the vessel contains 15 litres of ethanol

Substitute $x = 15$ and $t = 0$:

$15 = 30 + c\mathrm{e}^{-\frac{1}{10}\times 0}$

$\Rightarrow c = -15$

$\therefore x = 30 - 15\mathrm{e}^{-\frac{1}{10}t}$

In order for the ratio of ethanol to water to be $2:1$, we would need $\frac{2}{3} \times 40 = \frac{80}{3}$ litres of ethanol (to $\frac{1}{3} \times 40 = \frac{40}{3}$ litres of water).

Substitute $x = \frac{80}{3}$:

$\frac{80}{3} = 30 - 15\mathrm{e}^{-\frac{1}{10}t}$

$\Rightarrow \mathrm{e}^{-\frac{1}{10}t} = \frac{2}{9}$

$\Rightarrow -\frac{1}{10}t = \ln\frac{2}{9}$

$\Rightarrow t = -10\ln\frac{2}{9}$

$= 15$ seconds to the nearest second

34) c)

The model predicts that the ratio 2:1 occurs after about 15 seconds.
Experimentally, it occurs after 18 seconds.
Since these values are different, the model is not fully valid. A possible reason is that the mixture may not be perfectly uniform at all times, even though it is being stirred.

35) a)

Net rate of change of volume $= 0.1 - 0.05 = 0.05$ litres per minute
Initial volume is 5 litres so:
$V(t) = 5 + 0.05t$

35) b)

The inflow concentration is $2 + 3\mathrm{e}^{-0.08t}$ mg per litre
Inflow volume rate is 0.1 litres per minute
So the rate in $= 0.1(2 + 3\mathrm{e}^{-0.08t})$ mg per minute
The concentration inside of the tank is $\frac{m}{V(t)}$ mg per litre
Outflow volume rate is 0.05 litres per minute
So the rate out $= 0.05 \times \frac{m}{V(t)} = 0.05 \times \frac{m}{5+0.05t}$
$= \frac{1}{20} \times \frac{m}{5+\frac{t}{20}} = \frac{m}{100+t}$
$\frac{\mathrm{d}m}{\mathrm{d}t} =$ rate in $-$ rate out
$\therefore \frac{\mathrm{d}m}{\mathrm{d}t} = 0.1(2 + 3\mathrm{e}^{-0.08t}) - \frac{m}{100+t}$
$\Rightarrow \frac{\mathrm{d}m}{\mathrm{d}t} + \frac{m}{100+t} = \frac{1}{10}(2 + 3\mathrm{e}^{-0.08t})$ as required

35) c)

$\frac{\mathrm{d}m}{\mathrm{d}t}+\frac{m}{100+t}=\frac{1}{10}(2+3\mathrm{e}^{-0.08t})$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{100+t}\mathrm{d}t}$

$=\mathrm{e}^{\ln|100+t|}$

$=|100+t|$

Let Integrating Factor $=(100+t)$

$\frac{\mathrm{d}m}{\mathrm{d}t}+\frac{m}{100+t}=\frac{1}{10}(2+3\mathrm{e}^{-0.08t})$

$\Rightarrow(100+t)\frac{\mathrm{d}m}{\mathrm{d}t}+m=\frac{1}{10}(100+t)(2+3\mathrm{e}^{-0.08t})$

$\Rightarrow\frac{\mathrm{d}}{\mathrm{d}t}\big((100+t)m\big)=\frac{1}{10}(100+t)(2+3\mathrm{e}^{-0.08t})$

Let $I=\int(100+x)(2+3\mathrm{e}^{-0.08x})\,\mathrm{d}x$

Use Integration by Parts:

$u=100+x \qquad \frac{\mathrm{d}v}{\mathrm{d}x}=2+3\mathrm{e}^{-0.08x}$

$\frac{\mathrm{d}u}{\mathrm{d}x}=1 \qquad v=2x-\frac{75}{2}\mathrm{e}^{-0.08x}$

$\therefore I=(100+x)\left(2x-\frac{75}{2}\mathrm{e}^{-0.08x}\right)-\int\left(2x-\frac{75}{2}\mathrm{e}^{-0.08x}\right)\mathrm{d}x$

$=(100+x)\left(2x-\frac{75}{2}\mathrm{e}^{-0.08x}\right)-\left(x^2+\frac{1875}{4}\mathrm{e}^{-0.08x}\right)+k$

$\Rightarrow(100+t)m=\frac{1}{10}\left((100+t)\left(2t-\frac{75}{2}\mathrm{e}^{-0.08t}\right)-\left(t^2+\frac{1875}{4}\mathrm{e}^{-0.08t}\right)\right)+c$

$\Rightarrow m=\frac{1}{10}\left(2t-\frac{75}{2}\mathrm{e}^{-0.08t}\right)+\frac{c-\left(t^2+\frac{1875}{4}\mathrm{e}^{-0.08t}\right)}{1000+10t}$

Initially the vessel contains 5 litres of pure water.

Substitute $m=0$ and $t=0$:

$0=\frac{1}{10}\left(2\times0-\frac{75}{2}\mathrm{e}^{-0.08\times0}\right)+\frac{c-\left(0^2+\frac{1875}{4}\mathrm{e}^{-0.08\times0}\right)}{1000+0}$

$\Rightarrow 0=-\frac{15}{4}+\frac{c-\frac{1875}{4}}{1000}$

$\Rightarrow \mathrm{c}-\frac{1875}{4}=3750$

$\Rightarrow c=\frac{16875}{4}$

$\therefore m=\frac{1}{10}\left(2t-\frac{75}{2}\mathrm{e}^{-0.08t}\right)+\frac{\frac{16875}{4}-\left(t^2+\frac{1875}{4}\mathrm{e}^{-0.08t}\right)}{1000+10t}$

$\Rightarrow m=\frac{1}{5}t-\frac{15}{4}\mathrm{e}^{-0.08t}+\frac{16875-4t^2-1875\mathrm{e}^{-0.08t}}{4000+40t}$

35) d)

Substitute $t = 25$ into the volume:

$V(25) = 5 + 0.05 \times 25 = 6.25$ litres

Substitute $t = 25$ into the mass:

$m|_{t=25} = \frac{1}{5} \times 25 - \frac{15}{4}\mathrm{e}^{-0.08\times25} + \frac{16875-4\times25^2-1875\mathrm{e}^{-0.08\times25}}{4000+40\times25}$

$= 5 - 0.5075073121\ldots + \frac{14121.24634\ldots}{5000}$

$= 7.316741957\ldots$

So Concentration $= \frac{m(25)}{V(25)} = \frac{7.316741957\ldots}{6.25}$

$= 1.17$ mg per litre

The actual figure was 1.52 mg per litre

This is quite different, a $\frac{1.52-1.17}{1.52} \times 100 \approx 23\%$ difference.

The model does not appear particularly accurate and appears to underestimate the concentration. A possible reason for this is that the pure water and the chemical solution are not perfectly mixed.

36)

$\frac{\mathrm{d}T}{\mathrm{d}t} + T = 4\mathrm{e}^t \cos t$

Integrating Factor

$= \mathrm{e}^{\int 1\mathrm{d}t}$

$= \mathrm{e}^t$

$\frac{\mathrm{d}T}{\mathrm{d}t} + T = 4\mathrm{e}^t \cos t$

$\Rightarrow \mathrm{e}^t \frac{\mathrm{d}T}{\mathrm{d}t} + T\mathrm{e}^t = 4\mathrm{e}^{2t} \cos t$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(T\mathrm{e}^t) = 4\mathrm{e}^{2t} \cos t$

Let $I = \int \mathrm{e}^{2x} \cos x \,\mathrm{d}x$

Use Integration by Parts:

$u = \cos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\sin x \qquad v = \frac{1}{2}\mathrm{e}^{2x}$

$\therefore I = \frac{1}{2}\mathrm{e}^{2x} \cos x + \int \frac{1}{2}\mathrm{e}^{2x} \sin x \,\mathrm{d}x$

Use Integration by Parts again:

$u = \sin x$ $\qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{2}\mathrm{e}^{2x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \cos x$ $\qquad v = \frac{1}{4}\mathrm{e}^{2x}$

$\therefore I = \frac{1}{2}\mathrm{e}^{2x}\cos x + \left(\frac{1}{4}\mathrm{e}^{2x}\sin x - \int \frac{1}{4}\mathrm{e}^{2x}\cos x \,\mathrm{d}x\right)$

$\Rightarrow I = \frac{1}{2}\mathrm{e}^{2x}\cos x + \frac{1}{4}\mathrm{e}^{2x}\sin x - \frac{1}{4}I$

$\Rightarrow \frac{5}{4}I = \frac{1}{2}\mathrm{e}^{2x}\cos x + \frac{1}{4}\mathrm{e}^{2x}\sin x + k_1$

$\Rightarrow I = \frac{2}{5}\mathrm{e}^{2x}\cos x + \frac{1}{5}\mathrm{e}^{2x}\sin x + k_2$

So:

$T\mathrm{e}^{t} = 4\left(\frac{2}{5}\mathrm{e}^{2t}\cos t + \frac{1}{5}\mathrm{e}^{2t}\sin t\right) + c$

$\Rightarrow T = \frac{8}{5}\mathrm{e}^{t}\cos t + \frac{4}{5}\mathrm{e}^{t}\sin t + c\mathrm{e}^{-t}$

When $t = 0$, $T = 0$:

$0 = \frac{8}{5}\mathrm{e}^{0}\cos 0 + \frac{4}{5}\mathrm{e}^{0}\sin 0 + c\mathrm{e}^{0}$

$\Rightarrow 0 = \frac{8}{5} + c$

$\Rightarrow c = -\frac{8}{5}$

$\therefore T = \frac{8}{5}\mathrm{e}^{t}\cos t + \frac{4}{5}\mathrm{e}^{t}\sin t - \frac{8}{5}\mathrm{e}^{-t}$

So $T|_{t=\pi} = \frac{8}{5}\mathrm{e}^{\pi}\cos\pi + \frac{4}{5}\mathrm{e}^{\pi}\sin\pi - \frac{8}{5}\mathrm{e}^{-\pi}$

$= -\frac{8}{5}\mathrm{e}^{\pi} - \frac{8}{5}\mathrm{e}^{-\pi}$

$= -\frac{8}{5}(\mathrm{e}^{\pi} + \mathrm{e}^{-\pi})°\mathrm{C}$

37) a)

Use Separation of Variables

$\frac{\mathrm{d}N}{\mathrm{d}t} = -kN$

$\Rightarrow \int \frac{1}{N}\mathrm{d}N = \int -k\,\mathrm{d}t$

$\Rightarrow \ln|N| = -kt + c$

$\Rightarrow N = \pm \mathrm{e}^{-kt+c}$

$\Rightarrow N = A\mathrm{e}^{-kt}$

When $t = 0$, $N = N_0$

$N_0 = A\mathrm{e}^0 \Rightarrow A = N_0$

$\therefore N = N_0\mathrm{e}^{-kt}$

When $t = 1$, $N = 0.88N_0$

$0.88N_0 = N_0\mathrm{e}^{-k}$

$\Rightarrow 0.88 = \mathrm{e}^{-k}$

$\Rightarrow -k = \ln 0.88$

$\Rightarrow k = -\ln 0.88 = 0.128$ (3 SF)

37) b)

$\frac{\mathrm{d}N}{\mathrm{d}t} = -kN + 30t + 60$

$\Rightarrow \frac{\mathrm{d}N}{\mathrm{d}t} = -0.128N + 30t + 60$

37) c)

First, rearrange into the form $\frac{\mathrm{d}N}{\mathrm{d}t} + \mathrm{P}(t)N = \mathrm{Q}(t)$:

$\frac{\mathrm{d}N}{\mathrm{d}t} + 0.128N = 30t + 60$

Integrating Factor

$= \mathrm{e}^{\int 0.128\mathrm{d}t}$

$= \mathrm{e}^{0.128t}$

$\frac{\mathrm{d}N}{\mathrm{d}t} + 0.128N = 30t + 60$

$\Rightarrow \mathrm{e}^{0.128t}\frac{\mathrm{d}N}{\mathrm{d}t} + 0.128\mathrm{e}^{0.128t}N = 30t\mathrm{e}^{0.128t} + 60\mathrm{e}^{0.128t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{0.128t}N) = 30t\mathrm{e}^{0.128t} + 60\mathrm{e}^{0.128t}$

Let $I = \int 30x\mathrm{e}^{0.128x}\,\mathrm{d}x$

Use Integration by Parts:

$u = 30x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{0.128x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 30 \qquad v = \frac{1}{0.128}\mathrm{e}^{0.128x}$

$\therefore I = 30x \times \frac{1}{0.128}\mathrm{e}^{0.128x} - \int 30 \times \frac{1}{0.128}\mathrm{e}^{0.128x}\,\mathrm{d}x$

$= \frac{1875x\mathrm{e}^{0.128x}}{8} - \int \frac{1875\mathrm{e}^{0.128x}}{8}\mathrm{d}x$

$= \frac{1875x\mathrm{e}^{0.128x}}{8} - \frac{1875}{8} \times \frac{1}{0.128}\mathrm{e}^{0.128x} + k$

$= \frac{1875x\mathrm{e}^{0.128x}}{8} - \frac{234375\mathrm{e}^{0.128x}}{128} + k$

$\therefore \mathrm{e}^{0.128t}N = \frac{1875t\mathrm{e}^{0.128t}}{8} - \frac{234375\mathrm{e}^{0.128t}}{128} + \frac{1875}{4}\mathrm{e}^{0.128t} + c$

$\Rightarrow \mathrm{e}^{0.128t}N = \frac{1875t\mathrm{e}^{0.128t}}{8} - \frac{174375\mathrm{e}^{0.128t}}{128} + c$

$\Rightarrow N = \frac{1875t}{8} - \frac{174375}{128} + c\mathrm{e}^{-0.128t}$

When $t = 0$, $N = 500$

$500 = \frac{1875}{8} \times 0 - \frac{174375}{128} + c\mathrm{e}^{0}$

$\Rightarrow c = \frac{238375}{128}$

$\therefore N = \frac{1875t}{8} - \frac{174375}{128} + \frac{238375}{128}\mathrm{e}^{-0.128t}$

Or $N = 234.375t - 1362.30 + 1862.30\mathrm{e}^{-0.128t}$

37) d)

Possible limitations include:

- The model assumes the leaving rate is always proportional to N, but in reality customers may leave because of competition, pricing, or customer satisfaction.
- The model assumes that the marketing term $30t$ increases forever, which is unrealistic as the market is finite.
- The model assumes the partnerships adds exactly 60 subscribers per year continuously, which may fluctuate.
- The model ignores seasonal effects, and assumes subscribers join and leave continuously, rather than in discrete monthly renewals.

38) a)

$\left(\frac{4+2t}{\sqrt{t}}\right)\frac{\mathrm{d}M}{\mathrm{d}t}+\frac{4M}{\sqrt{t}}=105(2+t)$

First, rearrange into the form $\frac{\mathrm{d}M}{\mathrm{d}t}+\mathrm{P}(t)M=\mathrm{Q}(t)$:

$(4+2t)\frac{\mathrm{d}M}{\mathrm{d}t}+4M=105(2+t)\sqrt{t}$

$\Rightarrow \frac{\mathrm{d}M}{\mathrm{d}t}+\frac{4M}{4+2t}=\frac{105(2+t)\sqrt{t}}{4+2t}$

$\Rightarrow \frac{\mathrm{d}M}{\mathrm{d}t}+\frac{2M}{2+t}=\frac{105(2+t)\sqrt{t}}{2(2+t)}$

$\Rightarrow \frac{\mathrm{d}M}{\mathrm{d}t}+\frac{2M}{2+t}=\frac{105}{2}\sqrt{t}$

Integrating Factor

$=\mathrm{e}^{\int\frac{2}{2+t}\mathrm{d}t}$

$=\mathrm{e}^{2\ln|2+t|}$

$=\mathrm{e}^{\ln|(2+t)^2|}$

$=|(2+t)^2|$

Let Integrating Factor $=(2+t)^2$

$\frac{\mathrm{d}M}{\mathrm{d}t}+\frac{2M}{2+t}=\frac{105}{2}\sqrt{t}$

$\Rightarrow (2+t)^2\frac{\mathrm{d}M}{\mathrm{d}t}+\frac{2M}{2+t}\times(2+t)^2=\frac{105}{2}(2+t)^2\sqrt{t}$

$\Rightarrow (2+t)^2\frac{\mathrm{d}M}{\mathrm{d}t}+2(2+t)M=\frac{105}{2}(2+t)^2\sqrt{t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}((2+t)^2M)=\frac{105}{2}(2+t)^2\sqrt{t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}((2+t)^2M)=\frac{105}{2}(4+4t+t^2)t^{\frac{1}{2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}((2+t)^2 M) = 210t^{\frac{1}{2}} + 210t^{\frac{3}{2}} + \frac{105}{2}t^{\frac{5}{2}}$

$\Rightarrow (2+t)^2 M = 140t^{\frac{3}{2}} + 84t^{\frac{5}{2}} + 15t^{\frac{7}{2}} + c$

$\Rightarrow M = \frac{140t^{\frac{3}{2}} + 84t^{\frac{5}{2}} + 15t^{\frac{7}{2}} + c}{(2+t)^2}$

When $t = 0$, $M = 4$:

$4 = \frac{140\times 0^{\frac{3}{2}} + 84\times 0^{\frac{5}{2}} + 15\times 0^{\frac{7}{2}} + c}{(2+0)^2}$

$\Rightarrow 4 = \frac{c}{4}$

$\Rightarrow c = 16$

$\therefore M = \frac{140t^{\frac{3}{2}} + 84t^{\frac{5}{2}} + 15t^{\frac{7}{2}} + 16}{(2+t)^2}$

When $t = 6$:

$M|_{t=6} = \frac{140\times 6^{\frac{3}{2}} + 84\times 6^{\frac{5}{2}} + 15\times 6^{\frac{7}{2}} + 16}{(2+6)^2}$

$= \frac{17417.17513\ldots}{64}$

$= 272$ grams (3 SF)

38) b)

$\left(\frac{4+2t}{\sqrt{t}}\right)\frac{\mathrm{d}M}{\mathrm{d}t} + \frac{4M}{\sqrt{t}} = 105(2+t)$

$\Rightarrow \left(\frac{4+2t}{\sqrt{t}}\right)\frac{\mathrm{d}M}{\mathrm{d}t} = 105(2+t) - \frac{4M}{\sqrt{t}}$

$\Rightarrow \frac{\mathrm{d}M}{\mathrm{d}t} = \left(105(2+t) - \frac{4M}{\sqrt{t}}\right)\left(\frac{\sqrt{t}}{4+2t}\right)$

When $t = 3$:

$M|_{t=3} = \frac{140\times 3^{\frac{3}{2}} + 84\times 3^{\frac{5}{2}} + 15\times 3^{\frac{7}{2}} + 16}{(2+3)^2}$

$= \frac{2754.372327\ldots}{25}$

$= 110.1748931\ldots$ grams

Then:

$\left.\frac{\mathrm{d}M}{\mathrm{d}t}\right|_{t=3} = \left(105(2+3) - \frac{4\times 110.1748931\ldots}{\sqrt{3}}\right)\left(\frac{\sqrt{3}}{4+2\times 3}\right)$

$= 46.9$ grams per hour (3 SF)

38) c)

The model assumes that the impurity is evenly mixed throughout the system at all times, which may not be true in practice.

39)

We are told the rate of decrease is proportional to $x^2 + 16$, so

$\frac{\mathrm{d}x}{\mathrm{d}t} = -k(x^2 + 16)$ where $k > 0$

Separate the variables:

$\int \frac{1}{x^2+16}\,\mathrm{d}x = \int -k\,\mathrm{d}t$

$\Rightarrow \int \frac{1}{16\left(1+\frac{1}{16}x^2\right)}\,\mathrm{d}x = \int -k\,\mathrm{d}t$

$\Rightarrow \frac{1}{16}\int \frac{1}{1+\left(\frac{1}{4}x\right)^2}\,\mathrm{d}x = \int -k\,\mathrm{d}t$

$\Rightarrow \frac{1}{16} \times \frac{1}{\frac{1}{4}} \times \arctan\left(\frac{1}{4}x\right) = -kt + c$

$\Rightarrow \frac{1}{4}\arctan\left(\frac{1}{4}x\right) = -kt + c$

Substitute $x = 120$ and $t = 0$:

$\frac{1}{4}\arctan\left(\frac{1}{4} \times 120\right) = -k \times 0 + c$

$\Rightarrow c = \frac{1}{4}\arctan 30$

Substitute $x = 60$ and $t = 50$:

$\frac{1}{4}\arctan\left(\frac{1}{4} \times 60\right) = -50k + c$

$\Rightarrow \frac{1}{4}\arctan 15 = -50k + \frac{1}{4}\arctan 30$

$\Rightarrow k = \frac{1}{200}\arctan 30 - \frac{1}{200}\arctan 15$

$\Rightarrow k = \frac{1}{200}(\arctan 30 - \arctan 15)$

$\therefore \frac{1}{4}\arctan\left(\frac{1}{4}x\right) = -\frac{1}{200}(\arctan 30 - \arctan 15)t + \frac{1}{4}\arctan 30$

$\Rightarrow \arctan\left(\frac{1}{4}x\right) = \frac{1}{50}(\arctan 15 - \arctan 30)t + \arctan 30$

$\Rightarrow x = 4\tan\left(\frac{1}{50}(\arctan 15 - \arctan 30)t + \arctan 30\right)$

When $t = 200$:

$x|_{t=200} = 4\tan\left(\frac{1}{50}(\arctan 15 - \arctan 30) \times 200 + \arctan 30\right)$

$= 23.8$ mg (3 SF)

40) a)

Driving Force $= (6 + 2t)$ N

Resistive Force $= 3v$ N

Mass $= 3$ kg

Acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$

$F = ma$

$\Rightarrow (6 + 2t) - 3v = 3\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = 2 + \frac{2}{3}t - v$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + v = 2 + \frac{2}{3}t$

40) b)

$\frac{\mathrm{d}v}{\mathrm{d}t} + v = 2 + \frac{2}{3}t$

Integrating Factor

$= \mathrm{e}^{\int 1 \mathrm{d}t}$

$= \mathrm{e}^{t}$

$\frac{\mathrm{d}v}{\mathrm{d}t} + v = 2 + \frac{2}{3}t$

$\Rightarrow \mathrm{e}^{t}\frac{\mathrm{d}v}{\mathrm{d}t} + \mathrm{e}^{t}v = 2\mathrm{e}^{t} + \frac{2}{3}t\mathrm{e}^{t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{t}v) = 2\mathrm{e}^{t} + \frac{2}{3}t\mathrm{e}^{t}$

Let $I = \int \frac{2}{3}x\mathrm{e}^{x}\,\mathrm{d}x$

Use Integration by Parts:

$u = \frac{2}{3}x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3}$ $\qquad$ $v = \mathrm{e}^{x}$

$\therefore I = \frac{2}{3}x\mathrm{e}^{x} - \int \frac{2}{3}\mathrm{e}^{x}\,\mathrm{d}x$

$= \frac{2}{3}x\mathrm{e}^{x} - \frac{2}{3}\mathrm{e}^{x} + k$

So:

$\mathrm{e}^{t}v = 2\mathrm{e}^{t} + \frac{2}{3}t\mathrm{e}^{t} - \frac{2}{3}\mathrm{e}^{t} + c$

$\Rightarrow \mathrm{e}^{t}v = \frac{4}{3}\mathrm{e}^{t} + \frac{2}{3}t\mathrm{e}^{t} + c$

$\Rightarrow v = \frac{4}{3} + \frac{2}{3}t + c\mathrm{e}^{-t}$

When $t = 0$, $v = 0$:

$0 = \frac{4}{3} + \frac{2}{3} \times 0 + c\mathrm{e}^{0}$

$\Rightarrow c = -\frac{4}{3}$

$\therefore v = \frac{4}{3} + \frac{2}{3}t - \frac{4}{3}\mathrm{e}^{-t}$

40) c)

When $t = 3$:

$v|_{t=3} = \frac{4}{3} + \frac{2}{3} \times 3 - \frac{4}{3}\mathrm{e}^{-3}$

$= \left(\frac{10}{3} - \frac{4}{3}\mathrm{e}^{-3}\right)$ ms^{-1}

40) d)

Driving Force $= 9$ N

Resistive Force $= 3v$ N

Mass $= 3$ kg

Acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$

$F = ma$

$\Rightarrow 9 - 3v = 3\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = 3 - v$

40) e)

$\frac{\mathrm{d}v}{\mathrm{d}t} = 3 - v$

Method 1: Use an Integrating Factor

First, rearrange into the form $\frac{\mathrm{d}v}{\mathrm{d}t} + \mathrm{P}(t)v = \mathrm{Q}(t)$:

$\frac{\mathrm{d}v}{\mathrm{d}t} + v = 3$

Integrating Factor

$= \mathrm{e}^{\int 1 \mathrm{d}t}$

$= \mathrm{e}^{t}$

$\frac{\mathrm{d}v}{\mathrm{d}t} + v = 3$

$\Rightarrow \mathrm{e}^{t}\frac{\mathrm{d}v}{\mathrm{d}t} + \mathrm{e}^{t}v = 3\mathrm{e}^{t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{t}v) = 3\mathrm{e}^{t}$

$\Rightarrow \mathrm{e}^{t}v = 3\mathrm{e}^{t} + c$

$\Rightarrow v = 3 + c\mathrm{e}^{-t}$

When $t = 3$, $v = \frac{10}{3} - \frac{4}{3}\mathrm{e}^{-3}$:

$\frac{10}{3} - \frac{4}{3}\mathrm{e}^{-3} = 3 + c\mathrm{e}^{-3}$

$\Rightarrow \frac{10}{3}\mathrm{e}^{3} - \frac{4}{3} = 3\mathrm{e}^{3} + c$

$\Rightarrow c = \frac{1}{3}\mathrm{e}^{3} - \frac{4}{3}$

$\therefore v = 3 + \left(\frac{1}{3}\mathrm{e}^{3} - \frac{4}{3}\right)\mathrm{e}^{-t}$

Method 2: Separation of Variables

$\int \frac{1}{3-v}\mathrm{d}v = \int 1\,\mathrm{d}t$

$\Rightarrow -\ln|3 - v| = t + c$

$\Rightarrow \ln|3 - v| = k - t$

$\Rightarrow 3 - v = \pm\mathrm{e}^{k-t}$

$\Rightarrow 3 - v = A\mathrm{e}^{-t}$

$\Rightarrow v = 3 - A\mathrm{e}^{-t}$

When $t = 3$, $v = \frac{10}{3} - \frac{4}{3}\mathrm{e}^{-3}$:

$\frac{10}{3} - \frac{4}{3}\mathrm{e}^{-3} = 3 + A\mathrm{e}^{-3}$

$\Rightarrow \frac{10}{3}\mathrm{e}^{3} - \frac{4}{3} = 3\mathrm{e}^{3} + A$

$\Rightarrow A = \frac{1}{3}\mathrm{e}^{3} - \frac{4}{3}$

$\therefore v = 3 + \left(\frac{1}{3}\mathrm{e}^{3} - \frac{4}{3}\right)\mathrm{e}^{-t}$

41) a)

Driving Force $= P = kt$ N

Resistive Force $= Q = cu$ N

Mass $= 0.4$ kg

Acceleration $a = \frac{\mathrm{d}u}{\mathrm{d}t}$

$F = ma$

$\Rightarrow kt - cu = 0.4\frac{\mathrm{d}u}{\mathrm{d}t}$

$\Rightarrow \frac{2}{5}\frac{\mathrm{d}u}{\mathrm{d}t} = kt - cu$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}t} = \frac{5}{2}(kt - cu)$

41) b)

$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{5}{2}(kt - cu)$

First, rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}t} + \mathrm{P}(t)u = \mathrm{Q}(t)$:

$\frac{\mathrm{d}u}{\mathrm{d}t} + \frac{5}{2}cu = \frac{5}{2}kt$

Integrating Factor

$= \mathrm{e}^{\int \frac{5}{2}c\,\mathrm{d}t}$

$= \mathrm{e}^{\frac{5}{2}ct}$

$\frac{\mathrm{d}u}{\mathrm{d}t} + \frac{5}{2}cu = \frac{5}{2}kt$

$\Rightarrow \mathrm{e}^{\frac{5}{2}ct}\frac{\mathrm{d}u}{\mathrm{d}t} + \frac{5}{2}c\mathrm{e}^{\frac{5}{2}ct}u = \frac{5}{2}kt\mathrm{e}^{\frac{5}{2}ct}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{e}^{\frac{5}{2}ct}u\right) = \frac{5}{2}kt\mathrm{e}^{\frac{5}{2}ct}$

Let $I = \int \frac{5}{2}kx\mathrm{e}^{\frac{5}{2}cx}\,\mathrm{d}x$

Use Integration by Parts:

$$u = \frac{5}{2}kx \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{\frac{5}{2}cx}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{5}{2}k \qquad v = \frac{2}{5c}\mathrm{e}^{\frac{5}{2}cx}$$

$\therefore I = \frac{k}{c}x\mathrm{e}^{\frac{5}{2}cx} - \int \frac{k}{c}\mathrm{e}^{\frac{5}{2}cx}\,\mathrm{d}x$

$= \frac{k}{c}x\mathrm{e}^{\frac{5}{2}cx} - \frac{2k}{5c^2}\mathrm{e}^{\frac{5}{2}cx} + c_1$

So:

$\mathrm{e}^{\frac{5}{2}ct}u = \frac{k}{c}t\mathrm{e}^{\frac{5}{2}ct} - \frac{2k}{5c^2}\mathrm{e}^{\frac{5}{2}ct} + c_2$

$\Rightarrow u = \frac{k}{c}t - \frac{2k}{5c^2} + c_2\mathrm{e}^{-\frac{5}{2}ct}$

When $t = 0$, $u = 0$:

$0 = \frac{k}{c} \times 0 - \frac{2k}{5c^2} + c_2\mathrm{e}^0$

$\Rightarrow c_2 = \frac{2k}{5c^2}$

$\therefore u = \frac{k}{c}t - \frac{2k}{5c^2} + \frac{2k}{5c^2}\mathrm{e}^{-\frac{5}{2}ct}$

41) c)

Substitute $k = 1.6$ and $c = 0.8$:

$u = \frac{1.6}{0.8}t - \frac{2\times1.6}{5\times0.8^2} + \frac{2\times1.6}{5\times0.8^2}\mathrm{e}^{-\frac{5}{2}\times0.8t}$

$\Rightarrow u = 2t - 1 + \mathrm{e}^{-2t}$

Acceleration:

$a = \frac{\mathrm{d}u}{\mathrm{d}t} = 2 - 2\mathrm{e}^{-2t}$

$a|_{t=2} = 2 - 2\mathrm{e}^{-2\times2}$

$= 1.96$ ms^{-2} (3 SF)

41) d)

Driving Force $= P = 1.6$ N

Resistive Force $= Q = 0.8u$ N

Mass $= 0.4$ kg

Acceleration $a = \frac{\mathrm{d}u}{\mathrm{d}t}$

$F = ma$

$\Rightarrow 1.6 - 0.8u = 0.4\frac{\mathrm{d}u}{\mathrm{d}t}$

$\Rightarrow \frac{2}{5}\frac{\mathrm{d}u}{\mathrm{d}t} = 1.6 - 0.8u$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}t} = 4 - 2u$

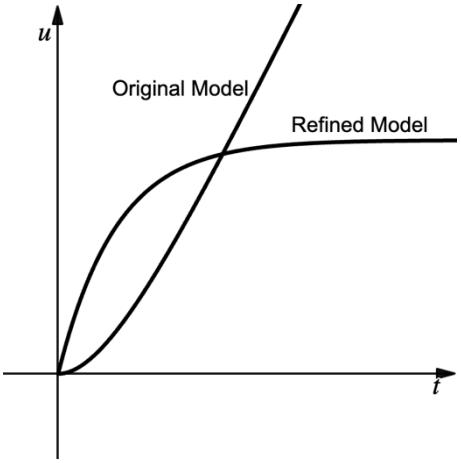
41) e)

The original model was $\frac{\mathrm{d}u}{\mathrm{d}t} = 4t - 2u$

$\Rightarrow u = 2t - 1 + \mathrm{e}^{-2t}$

The refined model is $\frac{\mathrm{d}u}{\mathrm{d}t} = 4 - 2u$

$\Rightarrow u = 2 - 2\mathrm{e}^{-2t}$



In the original model, the driving force continues to increase over time without bound. In the refined model, the diving force is constant, so the speed tends to a finite limiting value. So the speed predicted by the initial model will eventually exceed that of the refined model.

42) a)

Downwards is positive

Force downwards is just the weight $= mg$ N

Force upwards is the resistance $= km(v + 2)$ N

Mass $= m$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

$F = ma$:

$mg - km(v + 2) = m\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{g} - k(v + 2)$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{g} - kv - 2k$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + kv = \mathrm{g} - 2k$

42) b)

$\frac{dv}{dt} + kv = g - 2k$

Integrating Factor

$= e^{\int k dt}$

$= e^{kt}$

$\frac{dv}{dt} + kv = g - 2k$

$\Rightarrow e^{kt}\frac{dv}{dt} + ke^{kt}v = (g - 2k)e^{kt}$

$\Rightarrow \frac{d}{dt}(e^{kt}v) = (g - 2k)e^{kt}$

$\Rightarrow e^{kt}v = \frac{g-2k}{k}e^{kt} + c$

$\Rightarrow v = \frac{g-2k}{k} + ce^{-kt}$

When $t = 0$, $v = u$:

$u = \frac{g-2k}{k} + ce^{0}$

$\Rightarrow c = u - \frac{g-2k}{k}$

$\therefore v = \frac{g-2k}{k} + \left(u - \frac{g-2k}{k}\right)e^{-kt}$

42) c)

As $t \to \infty$, $e^{-kt} \to 0$

So $v \to \frac{g-2k}{k}$ ms^{-1}

So the terminal velocity $v_T = \frac{g-2k}{k}$ ms^{-1}

As long as $g > 2k$, otherwise the model suggests a negative terminal velocity, which would be inconsistent with downward motion.

42) d)

Downwards is positive

Force downwards is just the weight $= mg$ N

Force upwards is the resistance $= km\sqrt{v}$ N

Mass $= m$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

$F = ma$:

$mg - km\sqrt{v} = m\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{g} - k\sqrt{v}$

42) e)

The limiting velocity occurs when acceleration tends to 0, i.e. when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$

$\therefore \mathrm{g} - k\sqrt{v} = 0$

$\Rightarrow k\sqrt{v} = \mathrm{g}$

$\Rightarrow \sqrt{v} = \frac{\mathrm{g}}{k}$

$\Rightarrow v = \frac{\mathrm{g}^2}{k^2}$

We need to consider if the velocity tends to this value.

Consider if $v < \frac{\mathrm{g}^2}{k^2} \Rightarrow \sqrt{v} < \frac{\mathrm{g}}{k} \Rightarrow \mathrm{g} - k\sqrt{v} > 0 \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} > 0$ so the velocity is increasing when $v < \frac{\mathrm{g}^2}{k^2}$

Consider if $v > \frac{\mathrm{g}^2}{k^2} \Rightarrow \sqrt{v} > \frac{\mathrm{g}}{k} \Rightarrow \mathrm{g} - k\sqrt{v} < 0 \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} < 0$ so the velocity is decreasing when $v > \frac{\mathrm{g}^2}{k^2}$

So as the velocity increases when the particle is slower than $\frac{\mathrm{g}^2}{k^2}$ ms^{-1} and the velocity decreases when the particle is faster than $\frac{\mathrm{g}^2}{k^2}$ ms^{-1}, the velocity must tend towards $\frac{\mathrm{g}^2}{k^2}$ ms^{-1} as $t \to \infty$

43) a)

The carriage starts from rest at $t = 0$, so $v = 0$.

$\therefore$ the resistive force $= 0.12m \times 0 = 0$ N

43) b)

For $6 \leq t \leq 10$:

Engine Force $= 0$ N

Buffer Force $= 1.8m$ N

Resistance $= 0.12mv$ N

Mass $= m$ kg

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

$F = ma$:

$-1.8m - 0.12mv = m\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = -1.8 - 0.12v$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + 0.12v = -1.8$

43) c)

$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12v = -1.8$

Integrating Factor

$= \mathrm{e}^{\int 0.12\mathrm{d}t}$

$= \mathrm{e}^{0.12t}$

$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12v = -1.8$

$\Rightarrow \mathrm{e}^{0.12t}\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12\mathrm{e}^{0.12t}v = -1.8\mathrm{e}^{0.12t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{0.12t}v) = -1.8\mathrm{e}^{0.12t}$

$\Rightarrow \mathrm{e}^{0.12t}v = -15\mathrm{e}^{0.12t} + k_1$

$\Rightarrow v = -15 + k_1\mathrm{e}^{-0.12t}$

When $t = 10$, $v = 0$:

$0 = -15 + k_1\mathrm{e}^{-0.12\times 10}$

$\Rightarrow k_1 = 15\mathrm{e}^{1.2}$

$\therefore v = -15 + 15\mathrm{e}^{1.2} \times \mathrm{e}^{-0.12t}$

$\Rightarrow v = -15 + 15\mathrm{e}^{1.2-0.12t}$

43) d)

When $t = 6$:

$v|_{t=6} = -15 + 15e^{1.2-0.12\times6}$

$= -15 + 15e^{0.48}$

$= 9.24$ ms^{-1} (3 SF)

43) e)

For $0 \leq t \leq 6$:

Engine Force $= ct$ N

Resistance $= 0.12mv$ N

Mass $= m$ kg

Acceleration $= a = \frac{dv}{dt}$

$F = ma$:

$ct - 0.12mv = m\frac{dv}{dt}$

$\Rightarrow \frac{dv}{dt} = \frac{c}{m}t - 0.12v$

$\Rightarrow \frac{dv}{dt} + 0.12v = \frac{c}{m}t$

43) f)

$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12v = \frac{c}{m}t$

Integrating Factor

$= \mathrm{e}^{\int 0.12\mathrm{d}t}$

$= \mathrm{e}^{0.12t}$

$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12v = \frac{c}{m}t$

$\Rightarrow \mathrm{e}^{0.12t}\frac{\mathrm{d}v}{\mathrm{d}t} + 0.12\mathrm{e}^{0.12t}v = \frac{c}{m}t\mathrm{e}^{0.12t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{0.12t}v) = \frac{c}{m}t\mathrm{e}^{0.12t}$

Let $I = \frac{c}{m}x\mathrm{e}^{0.12x}$

Use Integration by Parts:

$$u = \frac{c}{m}x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{0.12x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{c}{m} \qquad v = \frac{25}{3}\mathrm{e}^{0.12x}$$

$\therefore I = \frac{25c}{3m}x\mathrm{e}^{0.12x} - \int \frac{25c}{3m}\mathrm{e}^{0.12x}\,\mathrm{d}x$

$= \frac{25c}{3m}x\mathrm{e}^{0.12x} - \frac{625c}{9m}\mathrm{e}^{0.12x} + k_2$

$\Rightarrow \mathrm{e}^{0.12t}v = \frac{25c}{3m}t\mathrm{e}^{0.12t} - \frac{625c}{9m}\mathrm{e}^{0.12t} + k_3$

$\Rightarrow v = \frac{25c}{3m}t - \frac{625c}{9m} + k_3\mathrm{e}^{-0.12t}$

When $t = 0$, $v = 0$:

$0 = \frac{25c}{3m} \times 0 - \frac{625c}{9m} + k_3\mathrm{e}^0$

$\Rightarrow k_3 = \frac{625c}{9m}$

$\therefore v = \frac{25c}{3m}t - \frac{625c}{9m} + \frac{625c}{9m}\mathrm{e}^{-0.12t}$

43) g)

When $t = 6$, the velocities must be the same.

From before: $v|_{t=6} = -15 + 15\mathrm{e}^{0.48}$

Also:

$$v|_{t=6} = \frac{25c}{3m} \times 6 - \frac{625c}{9m} + \frac{625c}{9m}\mathrm{e}^{-0.12\times 6}$$

$$= \frac{50c}{m} - \frac{625c}{9m} + \frac{625c}{9m}\mathrm{e}^{-0.72}$$

$$= -\frac{175c}{9m} + \frac{625c}{9m}\mathrm{e}^{-0.72}$$

$$= \frac{c}{m}\left(-\frac{175}{9} + \frac{625}{9}\mathrm{e}^{-0.72}\right)$$

Therefore:

$$-15 + 15\mathrm{e}^{0.48} = \frac{c}{m}\left(-\frac{175}{9} + \frac{625}{9}\mathrm{e}^{-0.72}\right)$$

$$\Rightarrow \frac{c}{m} = \frac{-15+15\mathrm{e}^{0.48}}{-\frac{175}{9}+\frac{625}{9}\mathrm{e}^{-0.72}}$$

$$= \frac{-135+135\mathrm{e}^{0.48}}{-175+625\mathrm{e}^{-0.72}}$$

$$= \frac{-27+27\mathrm{e}^{0.48}}{-35+125\mathrm{e}^{-0.72}}$$

$$= \frac{27\mathrm{e}^{0.48}-27}{125\mathrm{e}^{-0.72}-35}$$

43) h)

For $0 \leq t \leq 6$:

$v = \frac{25c}{3m}t - \frac{625c}{9m} + \frac{625c}{9m}\mathrm{e}^{-0.12t}$

$= \frac{c}{m}\left(\frac{25}{3}t - \frac{625}{9} + \frac{625}{9}\mathrm{e}^{-0.12t}\right)$

$s = \frac{c}{m}\int_0^6 \left(\frac{25}{3}t - \frac{625}{9} + \frac{625}{9}\mathrm{e}^{-0.12t}\right)\mathrm{d}t$

$= \frac{c}{m}\left[\frac{25}{6}t^2 - \frac{625}{9}t - \frac{15625}{27}\mathrm{e}^{-0.12t}\right]_0^6$

$= \frac{c}{m}\left(\left(\frac{25}{6}\times 6^2 - \frac{625}{9}\times 6 - \frac{15625}{27}\mathrm{e}^{-0.12\times 6}\right) - \left(\frac{25}{6}\times 0^2 - \frac{625}{9}\times 0 - \frac{15625}{27}\mathrm{e}^{0}\right)\right)$

$= \frac{c}{m}\left(\left(-\frac{800}{3} - \frac{15625}{27}\mathrm{e}^{-0.72}\right) - \left(-\frac{15625}{27}\right)\right)$

$= \frac{c}{m}\times 30.35170373\ldots$

$= \frac{27\mathrm{e}^{0.48}-27}{125\mathrm{e}^{-0.72}-35}\times 30.35170373\ldots$

$= 19.53528415\ldots$

For $6 \leq t \leq 10$:

$v = -15 + 15\mathrm{e}^{1.2-0.12t}$

$s = \int_6^{10}(-15 + 15\mathrm{e}^{1.2-0.12t})\,\mathrm{d}t = [-15t - 125\mathrm{e}^{1.2-0.12t}]_6^{10}$

$= (-15\times 10 - 125\mathrm{e}^{1.2-0.12\times 10}) - (-15\times 6 - 125\mathrm{e}^{1.2-0.12\times 6})$

$= (-150 - 125\mathrm{e}^{0}) - (-90 - 125\mathrm{e}^{0.48})$

$= -275 + 90 + 125\mathrm{e}^{0.48}$

$= -185 + 125\mathrm{e}^{0.48}$

$= 17.00930027\ldots$

Total distance $= 19.53528415\ldots + 17.00930027 + \cdots$

$= 37$ m to the nearest metre

44) a)

The rate of increase being proportional to the difference between the limiting speed and the current speed means

$\frac{\mathrm{d}v}{\mathrm{d}t} = k(V - v)$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + kv = kV$

44) b)

Given $v = V(1 - \mathrm{e}^{-kt})$

$\Rightarrow v = V - V\mathrm{e}^{-kt}$

Differentiate:

$\frac{\mathrm{d}v}{\mathrm{d}t} = Vk\mathrm{e}^{-kt}$

Now find $k(V - v)$:

$k(V - v) = k\left(V - (V - V\mathrm{e}^{-kt})\right) = k(V - V + V\mathrm{e}^{-kt}) = Vk\mathrm{e}^{-kt}$

$\therefore \frac{\mathrm{d}v}{\mathrm{d}t} = k(V - v)$

So it satisfies the differential equation.

Initial condition: when $t = 0$, $v = 0$:

$0 = V(1 - \mathrm{e}^{-k\times 0}) \Rightarrow 0 = V(1 - 1) = 0$

So it satisfies the initial condition.

Long-term behaviour:

As $t \to \infty$, $\mathrm{e}^{-kt} \to 0$, so $v \to V(1 - 0) = V$

So it satisfies the long-term behaviour

44) c)

$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2t}{1+t^2}v = \frac{2Vt}{1+t^2} + F(t)$

Integrating Factor

$= \mathrm{e}^{\int \frac{2t}{1+t^2}\mathrm{d}t}$

$= \mathrm{e}^{\ln|1+t^2|}$

$= |1 + t^2|$

Let Integrating Factor $= 1 + t^2$

44) d)

$F(t) = 0$:

$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2t}{1+t^2}v = \frac{2Vt}{1+t^2}$

$\Rightarrow (1+t^2)\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2t}{1+t^2}v \times (1+t^2) = \frac{2Vt}{1+t^2} \times (1+t^2)$

$\Rightarrow (1+t^2)\frac{\mathrm{d}v}{\mathrm{d}t} + 2tv = 2Vt$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left((1+t^2)v\right) = 2Vt$

$\Rightarrow (1+t^2)v = Vt^2 + c$

$\Rightarrow v = \frac{Vt^2+c}{1+t^2}$

Substitute $t = 0$ and $v = 0$:

$0 = \frac{V\times 0^2+c}{1+0^2}$

$\Rightarrow c = 0$

$\therefore v = \frac{Vt^2}{1+t^2}$

44) e)

Let $v = \frac{1}{2}V$:

$\frac{1}{2}V = \frac{Vt^2}{1+t^2}$

$\Rightarrow \frac{1}{2} = \frac{t^2}{1+t^2}$

$\Rightarrow 1 + t^2 = 2t^2$

$\Rightarrow 1 = t^2$

$\Rightarrow t = 1$

So it takes 1 hour $= 60$ minutes to reach half its limiting value.

44) f)

$F(t) = \mathrm{e}^{-t}$:

$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2t}{1+t^2}v = \frac{2Vt}{1+t^2} + \mathrm{e}^{-t}$

$\Rightarrow (1+t^2)\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2t}{1+t^2}v \times (1+t^2) = \frac{2Vt}{1+t^2} \times (1+t^2) + (1+t^2)\mathrm{e}^{-t}$

$\Rightarrow (1+t^2)\frac{\mathrm{d}v}{\mathrm{d}t} + 2tv = 2Vt + (1+t^2)\mathrm{e}^{-t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\big((1+t^2)v\big) = 2Vt + (1+t^2)\mathrm{e}^{-t}$

Let $I = \int (1+x^2)\mathrm{e}^{-x}\,\mathrm{d}x$

Use Integration by Parts:

$u = 1+x^2$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{-x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$ $\qquad$ $v = -\mathrm{e}^{-x}$

$\therefore I = -(1+x^2)\mathrm{e}^{-x} + \int 2x\mathrm{e}^{-x}\,\mathrm{d}x$

Use Integration by Parts again:

$u = 2x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{-x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2$ $\qquad$ $v = -\mathrm{e}^{-x}$

$\therefore I = -(1+x^2)\mathrm{e}^{-x} + (-2x\mathrm{e}^{-x} + \int 2\mathrm{e}^{-x}\,\mathrm{d}x)$

$= -(1+x^2)\mathrm{e}^{-x} - 2x\mathrm{e}^{-x} + \int 2\mathrm{e}^{-x}\,\mathrm{d}x$

$= -(1+x^2)\mathrm{e}^{-x} - 2x\mathrm{e}^{-x} - 2\mathrm{e}^{-x} + k_1$

$\therefore (1+t^2)v = Vt^2 - (1+t^2)\mathrm{e}^{-t} - 2t\mathrm{e}^{-t} - 2\mathrm{e}^{-t} + k$

$\Rightarrow v = \frac{Vt^2+k-\mathrm{e}^{-t}(t^2+2t+3)}{1+t^2}$

Substitute $t = 0$ and $v = 0$:

$0 = \frac{V\times 0^2+k-\mathrm{e}^{0}(0^2+2\times 0+3)}{1+0^2}$

$\Rightarrow 0 = \frac{k-3}{1}$

$\Rightarrow k = 3$

$\therefore v = \frac{Vt^2+3-\mathrm{e}^{-t}(t^2+2t+3)}{1+t^2}$

44) g)

We saw e) that the first model took 60 minutes to reach half-speed, so this is not very close to 34 minutes.

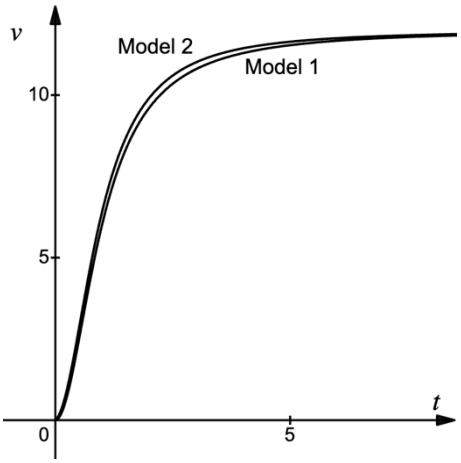
34 minutes $= \frac{34}{60} = \frac{17}{30}$ hours

Substitute $V = 12$ and $t = \frac{17}{30}$ into the first model:

$$v = \frac{12\times\left(\frac{17}{30}\right)^2}{1+\left(\frac{17}{30}\right)^2} = 2.9167\ldots$$

Substitute $V = 12$ and $t = \frac{17}{30}$ into the second model:

$$v = \frac{12\times\left(\frac{17}{30}\right)^2+3-e^{-\frac{17}{30}}\left(\left(\frac{17}{30}\right)^2+2\left(\frac{17}{30}\right)+3\right)}{1+\left(\frac{17}{30}\right)^2} = 3.2743\ldots$$



So the 2nd model gives a value that is closer to 6 (half the limiting speed), so the 2nd model is more consistent with this information.

45) a)

$t\frac{\mathrm{d}S}{\mathrm{d}t} = S - \frac{3t^2}{\sqrt{1+t^2}}$

First, rearrange into the form $\frac{\mathrm{d}S}{\mathrm{d}t} + \mathrm{P}(t)S = \mathrm{Q}(t)$:

$t\frac{\mathrm{d}S}{\mathrm{d}t} - S = -\frac{3t^2}{\sqrt{1+t^2}}$

$\Rightarrow \frac{\mathrm{d}S}{\mathrm{d}t} - \frac{1}{t}S = -\frac{3t}{\sqrt{1+t^2}}$

Integrating Factor

$= \mathrm{e}^{-\int \frac{1}{t}\mathrm{d}t}$

$= \mathrm{e}^{-\ln|t|}$

$= \mathrm{e}^{\ln|t^{-1}|}$

$= \left|\frac{1}{t}\right|$

Let Integrating Factor $= \frac{1}{t}$

$\frac{\mathrm{d}S}{\mathrm{d}t} - \frac{1}{t}S = -\frac{3t}{\sqrt{1+t^2}}$

$\Rightarrow \frac{1}{t}\frac{\mathrm{d}S}{\mathrm{d}t} - \frac{1}{t}S \times \frac{1}{t} = -\frac{3t}{\sqrt{1+t^2}} \times \frac{1}{t}$

$\Rightarrow \frac{1}{t}\frac{\mathrm{d}S}{\mathrm{d}t} - \frac{1}{t^2}S = -\frac{3}{\sqrt{1+t^2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{S}{t}\right) = -\frac{3}{\sqrt{1+t^2}}$

$\Rightarrow \frac{S}{t} = -3\,\mathrm{arsinh}\,t + c$

$\Rightarrow S = -3t\,\mathrm{arsinh}\,t + ct$

Substitute $S = 8$ and $t = 2$:

$8 = -3 \times 2 \times \mathrm{arsinh}\,2 + c \times 2$

$\Rightarrow 8 = -6\,\mathrm{arsinh}\,2 + 2c$

$\Rightarrow c = 4 + 3\,\mathrm{arsinh}\,2$

$\therefore S = -3t\,\mathrm{arsinh}\,t + (4 + 3\,\mathrm{arsinh}\,2)t$

45) b)

$S = 0$:

$-3t\,\mathrm{arsinh}\,t + (4 + 3\,\mathrm{arsinh}\,2)t = 0$

$\Rightarrow (-3\,\mathrm{arsinh}\,t + 4 + 3\,\mathrm{arsinh}\,2)t = 0$

$\Rightarrow t = 0$ or $-3\,\mathrm{arsinh}\,t + 4 + 3\,\mathrm{arsinh}\,2 = 0$

$\Rightarrow \mathrm{arsinh}\,t = \frac{4}{3} + \mathrm{arsinh}\,2$

$\Rightarrow t = \sinh\left(\frac{4}{3} + \mathrm{arsinh}\,2\right)$ seconds

45) c)

We want to find $\left.\frac{\mathrm{d}S}{\mathrm{d}t}\right|_{t=0}$

$t\frac{\mathrm{d}S}{\mathrm{d}t} = S - \frac{3t^2}{\sqrt{1+t^2}}$

$\Rightarrow \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{S}{t} - \frac{3t}{\sqrt{1+t^2}}$

$\Rightarrow \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{-3t\,\mathrm{arsinh}\,t+(4+3\,\mathrm{arsinh}\,2)t}{t} - \frac{3t}{\sqrt{1+t^2}}$

$\Rightarrow \frac{\mathrm{d}S}{\mathrm{d}t} = -3\,\mathrm{arsinh}\,t + (4+3\,\mathrm{arsinh}\,2) - \frac{3t}{\sqrt{1+t^2}}$

Then:

$\left.\frac{\mathrm{d}S}{\mathrm{d}t}\right|_{t=0} = -3\,\mathrm{arsinh}\,0 + (4+3\,\mathrm{arsinh}\,2) - \frac{3\times0}{\sqrt{1+0^2}}$

$= 4 + 3\,\mathrm{arsinh}\,2$

46) a)

Acceleration $= a = \frac{\mathrm{d}v}{\mathrm{d}t}$

Mass $= 2$ kg

$F = ma$:

$\frac{8}{t^2+2t+2} - \frac{2v}{(t+1)(t^2+2t+2)} = 2\frac{\mathrm{d}v}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{4}{t^2+2t+2} - \frac{v}{(t+1)(t^2+2t+2)}$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{(t+1)(t^2+2t+2)} = \frac{4}{t^2+2t+2}$

46) b)

$\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{v}{(t+1)(t^2+2t+2)}=\frac{4}{t^2+2t+2}$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{(t+1)(t^2+2t+2)}\mathrm{d}t}$

Partial Fractions:

$\frac{1}{(t+1)(t^2+2t+2)}\equiv\frac{A}{t+1}+\frac{Bt+C}{t^2+2t+2}$

$\Rightarrow 1\equiv A(t^2+2t+2)+(Bt+C)(t+1)$

Let $t=-1$: $1=A\Rightarrow A=1$

Let $t=0$: $1=1\times 2+C\times 1\Rightarrow C=-1$

Let $t=1$: $1=1\times 5+(B-1)\times 2\Rightarrow B-1=-2\Rightarrow B=-1$

$\therefore\frac{1}{(t+1)(t^2+2t+2)}\equiv\frac{1}{t+1}+\frac{-t-1}{t^2+2t+2}\equiv\frac{1}{t+1}-\frac{t+1}{t^2+2t+2}$

Back to Integrating Factor

$=\mathrm{e}^{\int\left(\frac{1}{t+1}-\frac{t+1}{t^2+2t+2}\right)\mathrm{d}t}$

$=\mathrm{e}^{\ln|t+1|-\frac{1}{2}\ln|t^2+2t+2|}$

$=\mathrm{e}^{\ln|t+1|-\ln\left|(t^2+2t+2)^{\frac{1}{2}}\right|}$

$=\mathrm{e}^{\ln\left|\frac{t+1}{(t^2+2t+2)^{\frac{1}{2}}}\right|}$

$=\left|\frac{t+1}{(t^2+2t+2)^{\frac{1}{2}}}\right|$

Let Integrating Factor $=\frac{t+1}{\sqrt{t^2+2t+2}}$

$\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{v}{(t+1)(t^2+2t+2)}=\frac{4}{t^2+2t+2}$

$\Rightarrow\frac{t+1}{\sqrt{t^2+2t+2}}\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{v}{(t+1)(t^2+2t+2)}\times\frac{t+1}{\sqrt{t^2+2t+2}}=\frac{4}{t^2+2t+2}\times\frac{t+1}{\sqrt{t^2+2t+2}}$

$\Rightarrow\frac{t+1}{\sqrt{t^2+2t+2}}\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{v}{(t^2+2t+2)^{\frac{3}{2}}}=\frac{4(t+1)}{(t^2+2t+2)^{\frac{3}{2}}}$

$\Rightarrow\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t+1}{\sqrt{t^2+2t+2}}\times v\right)=\frac{4(t+1)}{(t^2+2t+2)^{\frac{3}{2}}}$

Use Integration by Substitution:

Let $u=t^2+2t+2\Rightarrow\frac{\mathrm{d}u}{\mathrm{d}t}=2t+2\Rightarrow\mathrm{d}t=\frac{1}{2(t+1)}\mathrm{d}u$

$\int\frac{4(t+1)}{(t^2+2t+2)^{\frac{3}{2}}}\mathrm{d}t$

$= \int \frac{4(t+1)}{u^{\frac{3}{2}}} \times \frac{1}{2(t+1)} \mathrm{d}u$

$= \int \frac{2}{u^{\frac{3}{2}}} \mathrm{d}u$

$= \int 2u^{-\frac{3}{2}} \mathrm{d}u$

$= -4u^{-\frac{1}{2}} + c$

$= -4(t^2 + 2t + 2)^{-\frac{1}{2}} + c$

$\therefore \frac{t+1}{\sqrt{t^2+2t+2}} \times v = -4(t^2 + 2t + 2)^{-\frac{1}{2}} + c$

$\Rightarrow v = -\frac{4}{t+1} + \frac{c\sqrt{t^2+2t+2}}{t+1}$

$\Rightarrow v = \frac{c\sqrt{t^2+2t+2}-4}{t+1}$

Substitute $v = -2$ and $t = 0$:

$-2 = \frac{c\sqrt{0^2+2\times0+2}-4}{0+1}$

$\Rightarrow -2 = c\sqrt{2} - 4$

$\Rightarrow c = \frac{2}{\sqrt{2}} = \sqrt{2}$

$\therefore v = \frac{\sqrt{2}\sqrt{t^2+2t+2}-4}{t+1}$

$\Rightarrow v = \frac{\sqrt{2t^2+4t+4}-4}{t+1}$

Stationary means $v = 0$

$\Rightarrow 0 = \frac{\sqrt{2t^2+4t+4}-4}{t+1}$

$\Rightarrow 0 = \sqrt{2t^2 + 4t + 4} - 4$

$\Rightarrow \sqrt{2t^2 + 4t + 4} = 4$

$\Rightarrow 2t^2 + 4t + 4 = 16$

$\Rightarrow 2t^2 + 4t - 12 = 0$

$\Rightarrow t^2 + 2t - 6 = 0$

$\Rightarrow (t + 1)^2 - 1 - 6 = 0$

$\Rightarrow (t + 1)^2 = 7$

$\Rightarrow t = -1 \pm \sqrt{7}$

We can disregard $t = -1 - \sqrt{7}$ as $t \geq 0$

Check $-1 + \sqrt{7} \approx 1.64 < 2$ so it is in the required domain.

So the first time for which the particle is stationary is $t = -1 + \sqrt{7}$ seconds